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# From Deployment to Operation: A Reproducible IoT Testbed for Smart-Building HVAC with OTA and Observability

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***Abstract*—Smart-building research increasingly requires IoT platforms that support reproducible deployment and long-term operation under real conditions. This paper presents a field-operable and reproducible IoT testbed for smart-building HVAC, grounded in an architecture aligned with ISO/IEC 30141 and designed for zero-touch onboarding, fleet-wide Over-The-Air (OTA) lifecycle management, and end-to-end observability. The architecture integrates ESP32-based devices running FreeRTOS, multi-protocol communication using MQTT and CoAP, and a lightweight containerized backend with time-series persistence and monitoring.**

**The testbed was deployed in a university laboratory, where a rule-based HVAC control strategy reduced monthly energy consumption from 101.68 to 45.04 kWh (55.7%) while maintaining thermal comfort. Controlled experiments further evaluate protocol trade-offs, quantifying MQTT QoS impacts on actuation latency and comparing MQTT and CoAP for continuous telemetry under constrained conditions. In addition, the architecture enables progressive edge intelligence through data-driven model deployment without structural changes.**

**Rather than proposing new protocols or control algorithms, this work contributes a standards-aligned, observable, and evolvable IoT testbed, together with empirical evidence and practical design insights for field-oriented smart-building research.**

***Index Terms*—IoT testbed, smart buildings, HVAC, OTA, observability, MQTT, CoAP, reproducibility, edge intelligence**

## I. Introduction

With the advancement of smart technologies, human–computer interaction has become pervasive, increasing the demand for solutions focused on environmental monitoring, automation, and user comfort [1], [2]. In the context of *smart buildings*, such solutions are essential not only to improve occupant well-being but also to reduce operational costs and enhance energy efficiency.

This work presents an architectural extension grounded in the ISO/IEC 30141 reference architecture, instantiated and validated through a low-cost IoT (*Internet of Things*) testbed based on microcontrollers and multi-protocol communication. The proposed architecture targets the integration of heterogeneous sensors and actuators for smart-building environments while explicitly addressing operational concerns frequently overlooked in the literature, namely observability, automatic device configuration, and fleet-wide Over-The-Air (OTA) updates.

Despite the maturity of IoT technologies, the efficient and low-cost integration of heterogeneous devices remains a barrier to large-scale adoption [3]–[5]. Addressing this challenge requires not only appropriate hardware and communication protocols, but also cohesive architectural integration across firmware, backend services, and user-facing interfaces, ensuring interoperability and long-term maintainability. Rather than proposing new control algorithms or communication protocols, this work focuses on the architectural structuring and operational deployment of an IoT platform aligned with established standards. The central research question is how a complete IoT stack can be architecturally structured, deployed, updated, observed, and empirically evaluated in the field in a reproducible manner, and what practical trade-offs arise from protocol and management choices.

To this end, we report results from a real HVAC deployment in a laboratory environment, including a well-defined baseline for energy consumption and controlled experiments evaluating communication latency and reliability under different protocol configurations. Although the control logic is rule-based, the emphasis lies on deployment reproducibility, operational observability, and empirical evidence supporting architectural and operational design decisions faced by practitioners in smart-building IoT systems. The novelty of this work does not lie in proposing new communication protocols or control algorithms; instead, it resides in the *operational integration and experimental validation* of a complete IoT stack that is simultaneously self-configuring, observable, remotely updatable, and empirically evaluated under real field conditions.

While most existing smart-building platforms are evaluated either in laboratory settings or through partial simulations, often focusing on isolated technologies and lacking an integrated architecture that functionally unites these elements [6], this work demonstrates how architectural structuring, protocol selection, firmware lifecycle management, monitoring, and actuation reliability interact in a long-running deployment. By exposing these interactions through controlled experiments and

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real measurements, the paper advances practical knowledge on how design decisions affect reliability, reproducibility, and maintainability in smart-building IoT systems. Rather than focusing on novel control algorithms, this work positions the proposed architecture as the primary research artifact, resulting in a reproducible IoT testbed with operational observability and empirically grounded design trade-offs in real smart-building environments.

### A. Contributions

This work makes the following contributions:

- **Field-operable IoT testbed architecture**: Design and real-world deployment of a multi-protocol, self-configuring, and OTA-enabled smart-building IoT architecture.
- **End-to-end operational stack**: Integration of firmware, backend services, monitoring, and lifecycle management mechanisms supporting scalable device fleet operation.
- **Experimental HVAC deployment**: A real-campus case study demonstrating substantial HVAC energy reduction while preserving thermal comfort.
- **Protocol-level operational insights**: Empirical evaluation of MQTT QoS selection for critical actuation and comparative analysis of MQTT and CoAP under constrained IoT conditions.
- **Reproducible research artifacts**: Release of anonymized data schemas and a minimal deployment stack to support replication and future extensions toward edge intelligence.

## II. Related Work

This work is positioned at the intersection of four research strands: (i) IoT reference architectures and system integration, (ii) communication protocols for resource-constrained IoT devices, (iii) smart-building IoT testbeds, and (iv) operational support mechanisms such as observability and over-the-air (OTA) firmware management. While each of these topics has been extensively explored in isolation, their integration in reproducible, field-operable smart-building deployments remains limited.

Several studies have proposed reference architectures to address the complexity and heterogeneity inherent to IoT systems. Standardization efforts such as ISO/IEC 30141 establish a high-level architectural framework by organizing IoT systems into functional domains, including device, communication, application, and operations and management. These reference models provide a conceptual foundation for interoperability and scalability but intentionally avoid prescribing concrete implementation strategies or deployment practices. In the context of smart buildings, existing architectures frequently emphasize specific aspects such as data acquisition pipelines, cloud-based analytics, or collaborative and distributed computing paradigms [7]. While these approaches highlight the importance of adaptability and multi-layer coordination, they often abstract away operational concerns related to device lifecycle management, firmware evolution, and long-term maintainability.

Recent works propose dynamic and collaborative IoT architectures that integrate edge, fog, and cloud resources to improve responsiveness and scalability [8]. However, these architectures are commonly evaluated through simulations or analytical models, with limited empirical validation in real deployments involving constrained devices and heterogeneous communication protocols. In smart-building scenarios, architectural proposals frequently focus on energy optimization or sensing accuracy, treating device configuration, monitoring, and updates as secondary concerns [9]. As a result, operational aspects such as observability, automated onboarding, and fleet-wide firmware updates are often assumed rather than explicitly designed and experimentally evaluated.

Communication protocols for constrained IoT devices have also been widely studied. The HTTP protocol is widely regarded as unsuitable for low-power IoT devices due to its verbose text-based format and synchronous request–response model, which incur significant processing and communication overhead [10], [11]. Empirical evaluations show that HTTP can consume up to four times more CPU resources than message-oriented alternatives such as MQTT and AMQP, while also exhibiting higher end-to-end latency [10], [11]. AMQP, originally designed for enterprise messaging systems, offers advanced features such as durable queues, routing, and transactions [12], [13]. However, its relatively large protocol header and architectural complexity make it ill-suited for severely constrained devices, and in practice no native AMQP client is available for the ESP32 family of microcontrollers. In contrast, MQTT and CoAP are specifically designed for constrained environments and have therefore become the de facto choices in IoT systems. Existing comparative studies primarily focus on throughput, latency, and energy consumption under synthetic workloads [11], while fewer works connect these protocol-level metrics to operational decisions in real deployments, such as actuator command reliability or comfort-preserving HVAC control.

Several IoT testbeds have been proposed to support experimentation with building automation and energy management [14]. Most of these platforms rely on MQTT as a single communication backbone due to its lightweight publish–subscribe model and ease of integration with heterogeneous devices and services [15]–[17]. Typical deployments range from small-scale laboratory environments to instrumented residential or academic buildings. While these testbeds demonstrate the feasibility of IoT-driven energy optimization, they often emphasize specific aspects such as consumption monitoring, demand response, or protocol interoperability. As a result, key operational concerns are frequently treated as secondary or left as future work, including automatic device configuration, fleet-wide firmware updates, and continuous infrastructure observability during long-term operation.

Firmware update mechanisms are increasingly recognized as essential for maintaining IoT deployments after installation. Recent approaches explore techniques such as WebAssembly-based updates to improve portability and update speed [18]. However, these solutions may introduce execution overheads



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that limit their suitability for microcontroller-based sensing and control tasks. Moreover, OTA mechanisms are often evaluated independently from the monitoring and control infrastructure, despite their tight coupling in real deployments.

*a) Discussion and gap analysis.:* Table I summarizes representative smart-building IoT testbeds and highlights the main gaps addressed by this work. In contrast to prior platforms, the proposed approach combines physical deployment, multi-protocol support, automatic configuration, OTA-based lifecycle management, and integrated monitoring within a single modular architecture, enabling reproducible field experimentation rather than isolated component evaluation.

**TABLE I:** *Comparison of representative smart-building IoT testbeds*

| Feature | R1 | R2 | R3 | R4 | R5 | R6 | R7 | This Work |
|---|---|---|---|---|---|---|---|---|
| Physical deployment | ✓ | – | ✓ | ✓ | – | – | – | ✓ |
| Scalability | – | ✓ | – | – | ✓ | – | ✓ | ✓ |
| Integrated monitoring | – | – | – | – | – | – | – | ✓ |
| Energy-aware control | – | ✓ | ✓ | ✓ | – | – | – | ✓ |
| Multi-protocol support | – | – | – | – | ✓ | – | ✓ | ✓ |
| End-device interoperability | – | – | – | – | – | ✓ | ✓ | ✓ |
| OTA support | – | – | – | – | – | – | ✓ | ✓ |

**References:** R1 [15], R2 [16], R3 [17], R4 [19], R5 [20], R6 [21], R7 [18].

Despite the maturity of individual components such as MQTT brokers, OTA mechanisms, and time-series databases, the literature still lacks experimentally grounded studies that analyze how these components interact when deployed together in long-running smart-building environments. Most existing testbeds either focus on a single aspect, such as energy monitoring or protocol throughput, or lack operational mechanisms required for sustained deployments, including fleet-wide updates, availability monitoring, and failure diagnosis.

In contrast, this work treats the IoT platform itself as the experimental artifact. Rather than optimizing isolated components, the proposed testbed enables empirical evaluation of protocol reliability, actuation latency, and operational trade-offs under real usage conditions. This systems-oriented perspective complements prior work by providing experimentally validated insights into how architectural and operational decisions affect reliability, reproducibility, and maintainability in smart-building IoT systems.

## III. System Design and Methodology

### A. Research Methodology

This research adopted a system-oriented and experimental methodology focused on the design and validation of an IoT architecture for smart-building environments. The study began with an analysis of existing IoT reference architectures and smart-building testbeds, aiming to identify architectural and operational gaps, particularly related to scalability, device lifecycle management, and observability.

Based on this analysis, suitable technologies and architectural patterns were selected to support the development of a low-cost and extensible IoT testbed. The implementation proceeded incrementally across architectural layers, starting from device and edge layers with firmware development and validation, followed by a fog layer responsible for data transport and integration, and culminating in cloud services for data processing, storage, and system management.

Finally, a user interface was developed to enable monitoring and control of the HVAC system, allowing the proposed architecture to be evaluated under real deployment conditions.

### B. Architectural Design

The research started with a systematic analysis of existing IoT reference architectures, with particular emphasis on standard-based models. Among the analyzed approaches, the ISO/IEC 30141 reference architecture was adopted as the primary conceptual foundation, as it provides a well-established functional decomposition of IoT systems into domains such as device, communication, application, and operations and management. Figure 1(A) illustrates the high-level structure defined by ISO/IEC 30141.

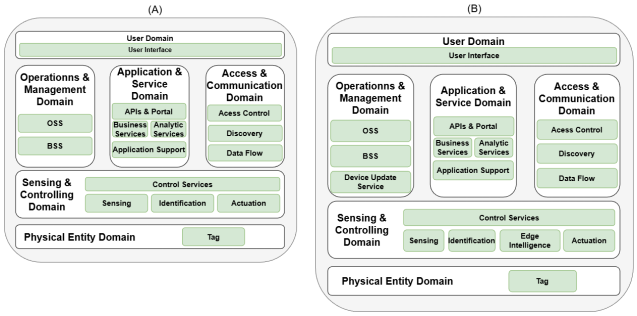



**Fig. 1:** *(A) ISO/IEC 30141 Reference Architecture. (B) Proposed architecture.*

Based on this reference model, this research proposes an **architectural extension** (Figure 1(B)) motivated by recent advances in the computational capabilities of IoT end devices. Modern microcontroller-based platforms are no longer limited to basic sensing and actuation tasks; instead, they are increasingly capable of executing local data processing and inference through **TinyML and lightweight embedded models**. This evolution enables the introduction of an additional intelligence layer within the sensing and control domain, allowing devices to locally analyze sensor data or collaboratively process information received from the edge layer.

Another central design decision concerns **scalability and long-term maintainability**, which are critical requirements in smart-building environments. To address these aspects, the architecture explicitly incorporates a **Device Update service** within the Operations and Management domain. This service is responsible for orchestrating **fleet-wide firmware updates and lightweight model deployment**, enabling versioning, rollback, and selective updates across heterogeneous end nodes.

By treating firmware and embedded models as first-class managed artifacts, the proposed architecture supports flexible reconfiguration of devices according to deployment context and user requirements. This approach increases the reliability and reproducibility of the testbed, allowing end nodes to dynamically assume different roles and behaviors without physical intervention.



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## IV. System Implementation

### A. Testbed Deployment and Experimental Setup

The testbed was developed using an incremental methodology combining architectural design, iterative implementation, and experimental validation in real environments. Hardware and software components were developed in parallel, enabling early integration and continuous validation under realistic operating conditions.

Experiments were conducted across multiple campus locations, with initial validation performed in two controlled laboratory environments, as shown in Figure 2. These rooms were selected due to their similar physical characteristics and independent HVAC units, allowing controlled experimentation while minimizing external interference. This approach ensured system stability and reproducibility before broader deployment.



**Fig. 2:** *Test rooms.*

### B. Implementation and Firmware Architecture

Table II summarizes the main technologies employed in the development and validation of the proposed IoT testbed.

**TABLE II:** *Technologies employed in the proposed IoT testbed*

| Layer / Component | Adopted Technology |
|---|---|
| User interface | Angular-based web application |
| Backend services | Spring Boot with WebFlux (reactive services) |
| Communication protocols | CoAP, MQTT, REST/HTTP |
| Database | PostgreSQL with TimescaleDB extension |
| Firmware framework | FreeRTOS with ESP-IDF |
| Hardware platform | ESP32 microcontroller |
| Sensors | Temperature, humidity, current, and light intensity |
| Actuators | Relay modules and HVAC units |

The firmware adopts a modular FreeRTOS-based architecture that enables concurrent sensing, communication, and control without blocking critical operations. Hardware abstraction is achieved through C++ interfaces, allowing sensors, actuators, ML models, and communication protocols to expose a uniform API. At startup, each device retrieves its configuration and operational parameters from the backend, also enabling version checks and OTA updates. The design prioritizes robustness, maintainability, and long-term reproducibility over device-specific optimization, supporting scalable deployment in the proposed testbed.

### C. Lifecycle Management and Edge Intelligence Services

Firmware maintenance in geographically distributed IoT deployments represents a major operational challenge. To eliminate the need for physical device access, an Over-The-Air (OTA) update mechanism was implemented to support centralized firmware lifecycle management. The OTA service compiles firmware binaries and exposes them through HTTP endpoints, allowing devices to periodically check for new versions and automatically apply updates. The update process follows a transactional procedure that ensures firmware replacement only after successful validation, as illustrated in Figure 3.

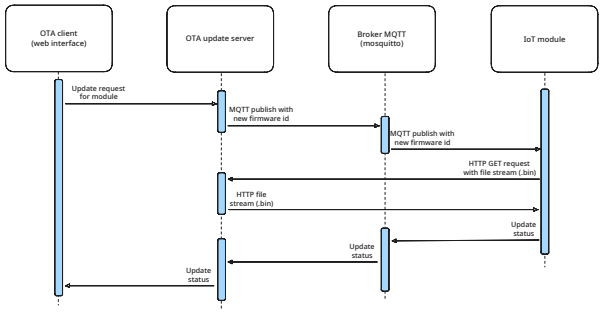



**Fig. 3:** *OTA update sequence.*

Beyond lifecycle management, the proposed architecture also enables the integration of edge-level intelligence capabilities. A key challenge in applying AI to smart-building environments is the limited availability of publicly accessible datasets related to occupancy, energy consumption, temperature, and humidity, mainly due to privacy and operational constraints. To address this limitation, a dataset of approximately ten million records was generated using data continuously collected by the deployed infrastructure. This was enabled by the end-to-end architecture of the proposed testbed, which supports continuous acquisition and centralized storage of device, edge, and cloud data streams, ensuring consistency and reproducibility.

Based on this dataset, a lightweight LSTM model was trained to predict indoor temperature and deployed for inference within the intelligent device layer of the proposed architecture, enabling edge-level intelligence directly on end nodes. This capability demonstrates the extensibility of the platform toward AI-driven control strategies while maintaining compatibility with resource-constrained devices.

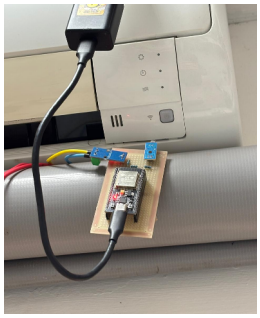
## V. Results and Analysis

### A. Case Study: HVAC Energy Reduction and Comfort Preservation

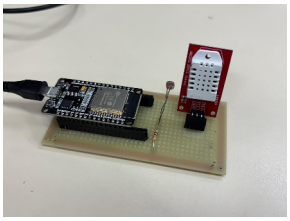
The proposed architecture was deployed in a university laboratory environment, selected for its controlled characteristics and suitability for evaluating smart-building automation strategies. The deployment integrated eight air-conditioning units equipped with ESP32-based actuator modules and environmental sensors for temperature, humidity, and luminosity, enabling continuous monitoring of comfort conditions and energy consumption. Figure 4 shows representative end nodes installed in the laboratory.



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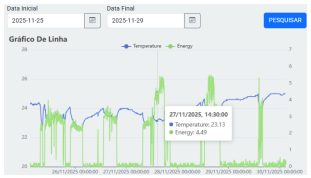


**(a)** *Actuation module with IR emitter and current sensing.*

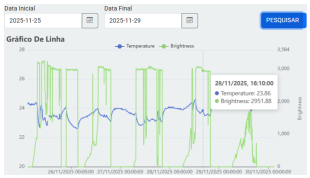


**(b)** *Environmental sensing module (temperature, humidity, luminosity).*

**Fig. 4:** *ESP32-based modules used in the HVAC deployment.*




**(a)** *Temperature vs. energy consumption*




**(b)** *Temperature vs. luminosity*

**Fig. 5:** *Thermal and operational behavior under automated HVAC control.*

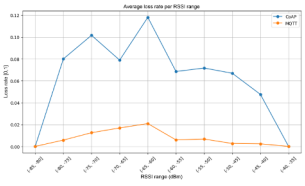
*a) Baseline and control strategy.:* Prior to automation, HVAC units followed a fixed operational schedule, remaining active during working hours regardless of room occupancy. This baseline configuration was monitored over a full monthly window, resulting in an average energy consumption of 101.68 kWh/month.

To reduce energy waste during unoccupied periods, a rule-based control strategy was implemented using luminosity as a proxy for occupancy. When luminosity readings remained below a predefined threshold for five consecutive minutes, the system automatically deactivated the HVAC unit via infrared actuation, resuming operation only when occupancy-related conditions were re-established.

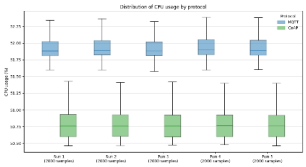
This strategy establishes a clear causal chain between environmental sensing, actuation, and physical effects. Luminosity readings trigger control decisions, actuation is confirmed through electric current measurements, and the resulting thermal response is observed through temperature variation, providing operational transparency and reproducible evaluation conditions.

*b) Energy and comfort behavior.:* Figure 5 illustrates the thermal and operational behavior of the environment under automated control. When the HVAC system is deactivated during periods of non-occupancy, energy consumption drops immediately, while temperature variations remain gradual and within acceptable comfort bounds (approximately $\pm 1\,^{\circ}\mathrm{C}$ around the configured setpoint). In contrast to the baseline configuration, energy consumption closely follows inferred occupancy patterns, indicating that savings are structurally induced by the control policy rather than incidental fluctuations.

*c) Observed impact.:* Under automated control, energy consumption averaged 45.04 kWh/month over a comparable monthly observation window, corresponding to a reduction of 55.7% relative to the baseline. Earlier exploratory analyses over shorter time intervals indicated reductions close to 52%; however, all quantitative results reported in this paper rely exclusively on full monthly aggregation windows, providing a more stable and representative assessment of energy consumption.




**(a)** *Packet loss over a 20,000-sample window.*




**(b)** *CPU utilization on ESP32 devices.*

**Fig. 6:** *MQTT vs. CoAP under constrained streaming conditions.*

*d) Control rules summary.:* The automation logic applied in this deployment can be summarized as follows:

(1) Activate HVAC only if ambient temperature exceeds $26\,^{\circ}\mathrm{C}$ **and** occupancy is detected.
(2) Deactivate HVAC when luminosity falls below a predefined threshold, indicating room vacancy.

These results demonstrate that even simple rule-based logic, when combined with robust deployment, monitoring, and actuation mechanisms, can yield substantial energy savings while preserving occupant comfort in real smart-building environments.

### B. QoS and Protocol Evaluation

This subsection evaluates communication reliability and performance under realistic deployment conditions, presenting a comparative analysis of MQTT and CoAP for continuous IoT data streaming under constrained environments.

*1) Protocols for IoT data streaming:* MQTT and CoAP were compared for continuous sensor data streaming under constrained conditions. MQTT was evaluated using QoS 0, while CoAP operated in non-confirmable (NON) mode to minimize overhead. Metrics included round-trip time (RTT), packet loss, CPU usage, heap variation, and RSSI.

Both protocols exhibited stable latency and low resource consumption. MQTT demonstrated greater robustness under unstable network conditions due to TCP-based retransmissions, while CoAP showed slightly lower CPU usage under stable links. Average CPU usage was $51.97 \pm 0.01\%$ for MQTT and $50.78 \pm 0.01\%$ for CoAP (95% confidence interval). These results confirm that the proposed architecture supports multi-protocol interoperability while maintaining predictable performance in real deployments.

## VI. Lessons Learned

The real-world deployment of the proposed IoT testbed provided several operational insights that are frequently underreported in laboratory-oriented studies. The lessons summarized



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below highlight practical aspects that directly influenced the reliability, scalability, and maintainability of the system.

**Lesson 1 — Reference architectures reduce design uncertainty.** Adopting a well-established reference architecture (e.g., ISO/IEC 30141) provides a structured starting point for IoT testbed design, improving architectural consistency and extensibility. Incremental deployment, starting with a small number of end nodes and progressively scaling after validation, proved essential to minimize early-stage integration risks.

**Lesson 2 — Simple control policies are suitable for infrastructure validation.** The rule-based HVAC control strategy was intentionally adopted to isolate infrastructure and deployment effects from model complexity. This approach enabled a clear evaluation of system reliability, observability, and operational behavior, establishing a reproducible baseline for future learning-based or adaptive control strategies.

**Lesson 3 — Indirect sensing can replace intrusive instrumentation.** Determining the operational status of HVAC units without hardware modification was initially challenging. Installing a current sensor on the power phase enabled indirect detection of the on/off state while simultaneously providing energy consumption measurements, proving effective for both monitoring and experimental validation.

**Lesson 4 — Continuous device observability is essential for long-term operation.** During early deployment phases, identifying whether ESP32 devices were online or experiencing connectivity issues was difficult. Integrating lightweight heartbeat-based monitoring using Zabbix enabled real-time visibility into device availability, substantially improving operational diagnostics and system reliability.

**Lesson 5 — Fleet-scale deployments require automated lifecycle management.** As the number of deployed devices increased, manual firmware updates became infeasible. The adoption of an OTA update mechanism enabled automated firmware distribution and version control across the fleet, supporting scalable and maintainable long-term operation without physical intervention.

**Lesson 6 — Dataset availability is a prerequisite for edge intelligence.** In smart-building environments, dataset availability remains a major barrier to the adoption of AI-based solutions due to privacy and operational constraints. Designing the architecture to support continuous data acquisition from the beginning of the deployment proved essential for enabling the construction of representative datasets and supporting future AI-driven control strategies.

## VII. Future Work

The proposed testbed enables several research directions that build upon the infrastructure and datasets already established. Future work includes extending the evaluation to multiple buildings and longer operational periods in order to assess scalability, seasonal effects, and heterogeneous occupancy patterns.

Another relevant direction is the integration of the platform with digital twin models, allowing what-if analyses and policy evaluation based on real operational data. The collected datasets also provide a foundation for investigating predictive monitoring and fault detection techniques, supporting more proactive maintenance and control strategies.

More advanced decision-making approaches, including learning-based methods, are left as future work and will be explored only after broader multi-building validation of the current rule-based control logic.

## VIII. Conclusion

This paper presented a field-operable and reproducible IoT testbed for smart-building HVAC deployments, grounded in an architecture aligned with ISO/IEC 30141. The proposed platform enables zero-touch onboarding, fleet-wide Over-The-Air (OTA) lifecycle management, and end-to-end observability, supporting long-term operation of distributed devices under real deployment conditions. A real-world HVAC deployment demonstrated that a simple rule-based control strategy reduced monthly energy consumption from 101.68 to 45.04 kWh (55.7%) while preserving thermal comfort, confirming that even lightweight control strategies can achieve substantial efficiency gains when combined with reliable sensing, actuation, and monitoring infrastructures. Protocol-level experiments further showed that MQTT and CoAP can be jointly supported with predictable latency and low resource consumption, exhibiting complementary robustness–overhead trade-offs that enable protocol selection according to application requirements without compromising system stability. Beyond monitoring and control, the architecture supports progressive integration of edge intelligence through large-scale data collection, model versioning, and over-the-air deployment mechanisms, allowing AI-driven functionalities to be incorporated without structural changes. Although evaluated in a single-campus environment, the architectural principles, deployment methodology, and operational mechanisms presented are transferable to other smart-building scenarios. By releasing an anonymized data schema and a minimal deployment stack, this work contributes reusable artifacts and empirical insights that support reproducible, field-oriented IoT research.

## Acknowledgements

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